

Лабораторная Работа №8. Настройка сетевых сервисов. DHCP

Администрирование локальных сетей

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Приобретение практических навыков по настройке динамического распределения IP-адресов посредством протокола DHCP (Dynamic Host Configuration Protocol) в локальной сети

Добавление сервера dns

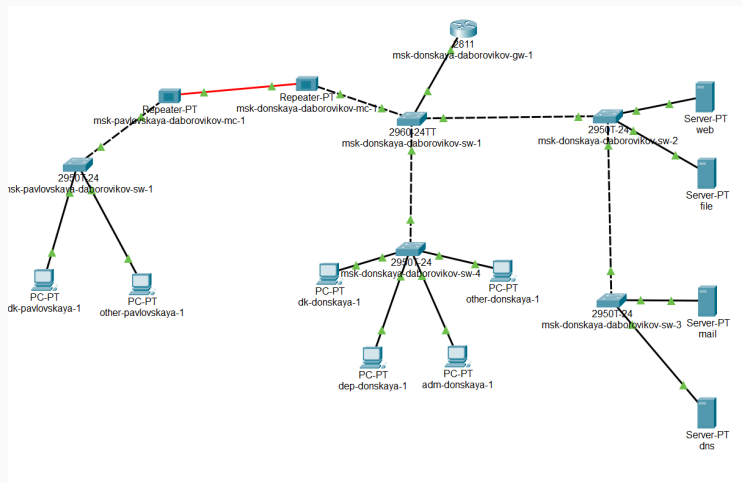
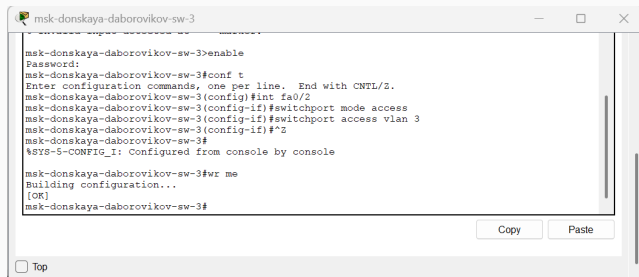


Figure 1: Добавление сервера dns

Активация порта на коммутаторе



```
msk-donskaya-daborovikov-sw-3>enable
Password:
msk-donskaya-daborovikov-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#int fa0/2
msk-donskaya-daborovikov-sw-3(config-if)#switchport mode access
msk-donskaya-daborovikov-sw-3(config-if)#switchport access vlan 3
msk-donskaya-daborovikov-sw-3(config-if)#^Z
msk-donskaya-daborovikov-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-3#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Copy Paste

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Figure 2: Активация порта на коммутаторе

Настройка конфигурации сервера

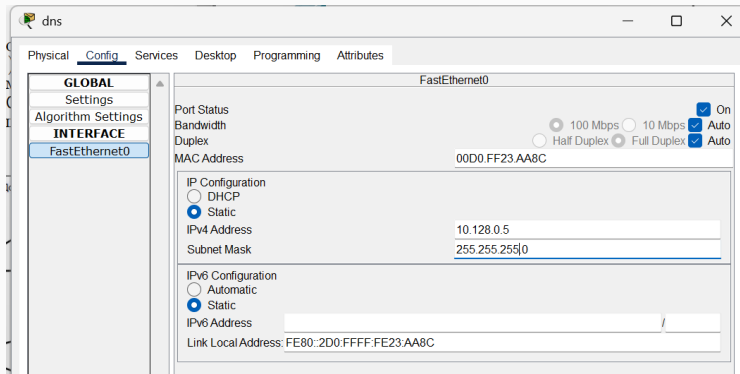


Figure 3: Настройка конфигурации сервера

Окно настройки сервиса DNS

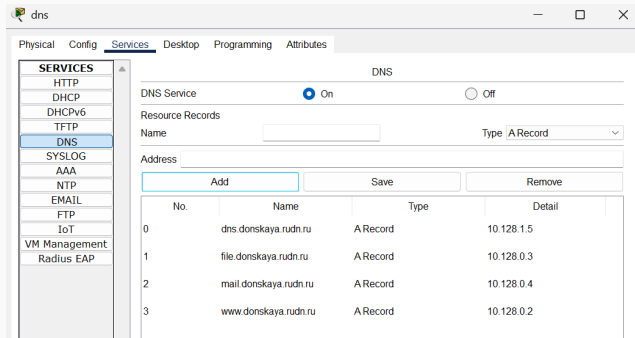


Figure 4: Окно настройки сервиса DNS

Настройка DHCP-сервиса на маршрутизаторе

```
msk-donskaya-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
Password:
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip name server 10.128.0.5
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-daborovikov-gw-1(config)#service dhcp
msk-donskaya-daborovikov-gw-1(config)#ip dhcp pool dx
msk-donskaya-daborovikov-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-daborovikov-gw-1(dhcp-config)#default-router 10.128.3.1
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-daborovikov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-daborovikov-gw-1(dhcp-config)#exit
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-daborovikov-gw-1(config)#ip dhcp pool departments
msk-donskaya-daborovikov-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-daborovikov-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-daborovikov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-daborovikov-gw-1(dhcp-config)#exit
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-daborovikov-gw-1(config)#ip dhcp pool adm
msk-donskaya-daborovikov-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-daborovikov-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-daborovikov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-daborovikov-gw-1(dhcp-config)#exit
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-daborovikov-gw-1(config)#ip dhcp pool ethw
msk-donskaya-daborovikov-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-daborovikov-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-daborovikov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-daborovikov-gw-1(dhcp-config)#exit
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-daborovikov-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-daborovikov-gw-1(config)#
```

Figure 5: Настройка DHCP-сервиса на маршрутизаторе, указание IP-адрес DNS-сервера; настройка DHCP; название конфигурируемому диапазону адресов (пулу адресов), указание адрес сети, шлюза и DNS-сервера; задание пулов адресов, исключаемых из динамического распределения

Динамическое распределения адресов

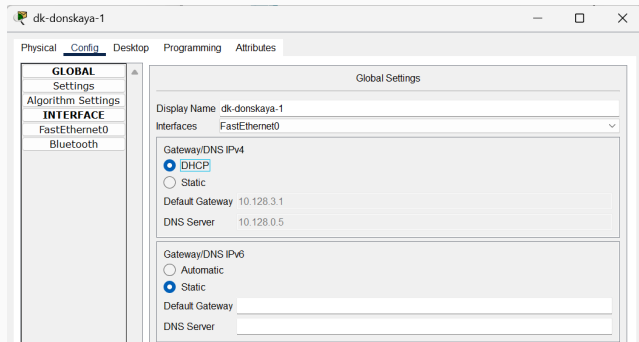
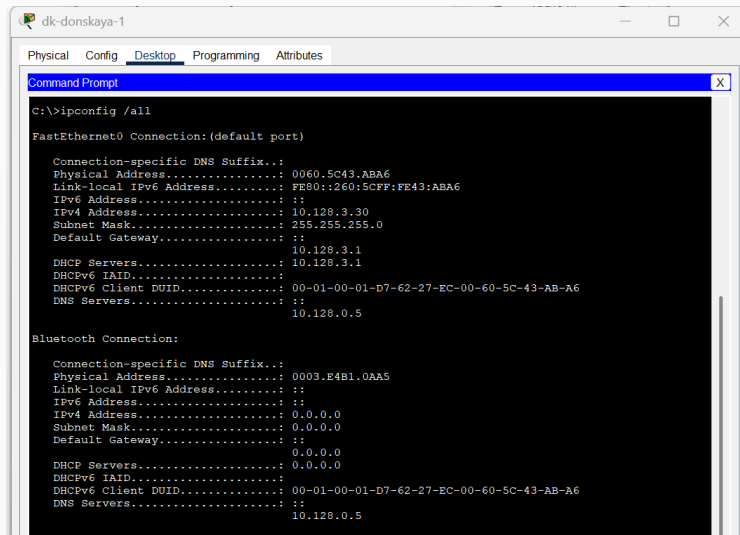


Figure 6: Замена статического распределения адресов на динамическое

Адреса оконечных устройств



```
dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt X

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

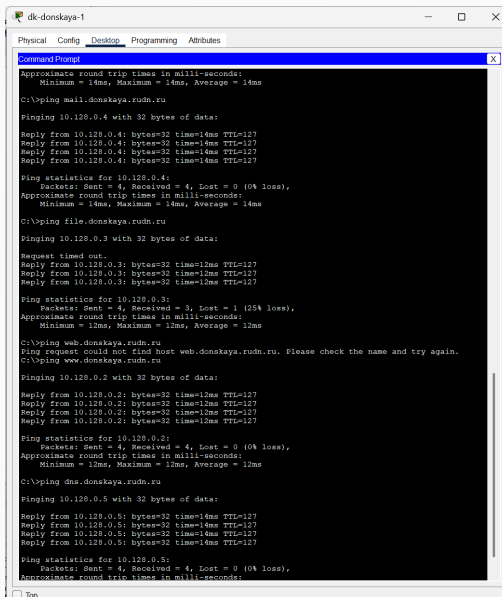
    Connection-specific DNS Suffix...: 
    Physical Address...: 0060.5C43.ABA6
    Link-local IPv6 Address...: FE80::260:5CFF:FE43:ABA6
    IPv6 Address...: ::
    IPv4 Address...: 10.128.3.30
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
                        10.128.3.1
    DHCP Servers...: 10.128.3.1
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-D7-62-27-EC-00-60-5C-43-AB-A6
    DNS Servers...: ::
                        10.128.0.5

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 0003.E4B1.0AA5
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
                        0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-D7-62-27-EC-00-60-5C-43-AB-A6
    DNS Servers...: ::
                        10.128.0.5
```

Figure 7: Адреса оконечных устройств

Доступность устройств из разных подсетей



```
dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
Approximate round trip times in milli-seconds:
  Minimum = 14ms, Maximum = 14ms, Average = 14ms

C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=14ms TTL=127
Reply from 10.128.0.4: bytes=32 time=14ms TTL=127
Reply from 10.128.0.4: bytes=32 time=14ms TTL=127
Reply from 10.128.0.4: bytes=32 time=14ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 14ms, Maximum = 14ms, Average = 14ms

C:\>ping file.donskaya.rudn.ru

Pinging 10.128.0.3 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.3: bytes=32 time=12ms TTL=127
Reply from 10.128.0.3: bytes=32 time=12ms TTL=127
Reply from 10.128.0.3: bytes=32 time=12ms TTL=127

Ping statistics for 10.128.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 12ms, Maximum = 12ms, Average = 12ms

C:\>ping web.donskaya.rudn.ru
Ping request could not find host web.donskaya.rudn.ru. Please check the name and try again.
C:\>ping www.donskaya.rudn.ru

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.2: bytes=32 time=12ms TTL=127
Reply from 10.128.0.2: bytes=32 time=12ms TTL=127
Reply from 10.128.0.2: bytes=32 time=12ms TTL=127
Reply from 10.128.0.2: bytes=32 time=12ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 12ms, Maximum = 12ms, Average = 12ms

C:\>ping dns.donskaya.rudn.ru

Pinging 10.128.0.5 with 32 bytes of data:

Reply from 10.128.0.5: bytes=32 time=14ms TTL=127
Reply from 10.128.0.5: bytes=32 time=14ms TTL=127
Reply from 10.128.0.5: bytes=32 time=14ms TTL=127
Reply from 10.128.0.5: bytes=32 time=14ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Запрос адреса по DHCP

The screenshot displays the Logical router interface with the Physical tab selected. The main window shows PDU Information at Device: dk-donskaya-1. The OSI Model tab is active, displaying Outbound PDU Details. The PDU is shown in a layered structure with In Layers and Out Layers. The Out Layers section highlights Layer 7: DNS. Below the layers, a list of layers is provided: Layer 7: DNS, Layer 6, Layer 5, Layer 4: UDP Src Port: 1034, Dst Port: 53, Layer 3: IP Header Src. IP: 10.128.3.30, Dest. IP: 10.128.0.5, Layer 2: Ethernet II Header 0060.5C43.ABA6 >> 00D0.9756.C701, and Layer 1: Port(s): FastEthernet0. A description at the bottom states: "1. The DNS client sends an A DNS query to the DNS server." Navigation buttons include "Challenge Me", "<< Previous Layer", and "Next Layer >>".

Simulation Panel

Vis	Time(sec)	Last Device	At Device	Type
	0.000	--	dk-donskaya-1	DNS
	0.001	dk-donskaya-1	msk-donskaya-daborovikov-sw-4	DNS
	0.002	msk-donskaya-daborovikov-sw-4	msk-donskaya-daborovikov-sw-1	DNS
	0.003	msk-donskaya-daborovikov-sw-1	msk-donskaya-daborovikov-gw-1	DNS
	0.004	msk-donskaya-daborovikov-gw-1	msk-donskaya-daborovikov-sw-1	DNS
	0.005	msk-donskaya-daborovikov-sw-1	msk-donskaya-daborovikov-sw-2	DNS
	0.006	msk-donskaya-daborovikov-sw-2	msk-donskaya-daborovikov-sw-3	DNS
	0.007	msk-donskaya-daborovikov-sw-3	dns	DNS
	0.008	dns	msk-donskaya-daborovikov-sw-3	DNS
	0.009	msk-donskaya-daborovikov-sw-3	msk-donskaya-daborovikov-sw-2	DNS
	0.010	msk-donskaya-daborovikov-sw-2	msk-donskaya-daborovikov-sw-1	DNS
	0.011	msk-donskaya-daborovikov-sw-1	msk-donskaya-daborovikov-gw-1	DNS
	0.012	msk-donskaya-daborovikov-gw-1	msk-donskaya-daborovikov-sw-1	DNS
	0.013	msk-donskaya-daborovikov-sw-1	msk-donskaya-daborovikov-sw-4	DNS
	0.014	msk-donskaya-daborovikov-sw-4	dk-donskaya-1	DNS
	0.014	--	dk-donskaya-1	ICMP

Figure 9: Запрос адреса по DHCP

В ходе выполнения лабораторной работы я приобрел практические навыки по настройке динамического распределения IP-адресов посредством протокола DHCP (Dynamic Host Configuration Protocol) в локальной сети